Taipei Veterans General Hospital

Department of Pathology and Laboratory Medicine
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## Sample Information

**Patient Name:** 郭秋華
**Gender:** Female
**ID No.:** A223064621
**History No.:** 5766450
**Age:** 61

**Ordering Doctor:** DOC3064F 陳育民
**Ordering REQ.:** D6MA7FF
**Signing in Date:** 2022/01/19

**Path No.:** S111-98207
**MP No.:** F22009
**Assay:** Oncomine Focus Assay
**Sample Type:** FFPE
**Block No.:** S111-00574A
**Percentage of tumor cells:** 50%

**Reporting Doctor:** DOC5466K 葉奕成 (Phone: 8#5466)

**Note:**

## Sample Cancer Type: Non-Small Cell Lung Cancer

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### Report Highlights

1 Relevant Biomarkers
6 Therapies Available
6 Clinical Trials

## Relevant Non-Small Cell Lung Cancer Variants

| Gene | Finding | Gene | Finding |
|---|---|---|---|
| ALK | ***EML4-ALK fusion*** | NTRK1 | None detected |
| BRAF | None detected | NTRK2 | None detected |
| EGFR | None detected | NTRK3 | None detected |
| ERBB2 | None detected | RET | None detected |
| KRAS | None detected | ROS1 | None detected |
| MET | None detected | | |

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.12(004).The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

## Relevant Biomarkers

| Tier | Genomic Alteration | Relevant Therapies (In this cancer type) | Relevant Therapies (In other cancer type) | Clinical Trials |
|---|---|---|---|---|
| IA | *EML4-ALK fusion*<br>EMAP like 4 - ALK receptor tyrosine kinase<br><br>**Prognostic significance:** None<br>**Diagnostic significance:** None | **alectinib** [1, 2]<br>**brigatinib** [1, 2]<br>**ceritinib** [1, 2]<br>**crizotinib** [1, 2]<br>**lorlatinib** [1, 2]<br>next-generation ALK inhibitor | brigatinib<br>ceritinib<br>crizotinib | 6 |

**Public data sources included in relevant therapies: FDA**[1], NCCN, **EMA**[2], ESMO
**Public data sources included in prognostic and diagnostic significance:** NCCN, ESMO
**Tier Reference:** Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

## Variant Details

### DNA Sequence Variants

| Gene | Amino Acid Change | Coding | Variant ID | Locus | Allele Frequency | Transcript | Variant Effect | Coverage |
|---|---|---|---|---|---|---|---|---|
| PDGFRA | p.(G313=) | c.939T>G | . | chr4:55133726 | 48.97% | NM_006206.6 | synonymous | 1997 |
| PDGFRA | p.(V824=) | c.2472C>T | . | chr4:55152040 | 99.55% | NM_006206.6 | synonymous | 1998 |
| FGFR4 | p.(P136L) | c.407C>T | . | chr5:176517797 | 99.75% | NM_213647.3 | missense | 2000 |
| EGFR | p.(Q787=) | c.2361G>A | . | chr7:55249063 | 49.60% | NM_005228.5 | synonymous | 2000 |
| RET | p.(L769=) | c.2307G>T | . | chr10:43613843 | 99.95% | NM_020975.6 | synonymous | 1996 |
| RET | p.(S904=) | c.2712C>G | . | chr10:43615633 | 60.28% | NM_020975.6 | synonymous | 1999 |

### Gene Fusions (RNA)

| Genes | Variant ID | Locus | Read Count |
|---|---|---|---|
| EML4-ALK | EML4-ALK.E13A20.COSF408.1 | chr2:42522656 - chr2:29446394 | 36178 |

## Biomarker Descriptions

**ALK (ALK receptor tyrosine kinase)**

Background: The ALK gene encodes the ALK receptor tyrosine kinase (RTK) with sequence similarity to the insulin receptor subfamily of kinases[1]. ALK is the target of recurrent alterations in cancer, the most common being chromosomal rearrangements that generate fusion genes containing the intact ALK tyrosine kinase domain combined with multiple partner genes[2]. ALK fusion kinases are constitutively activated and drive oncogenic transformation via activation of downstream STAT3, PI3K/AKT/MTOR, and RAS/RAF/MEK/ERK pathways[2,3,4,5].

Alterations and prevalence: ALK was discovered by positional cloning of translocations involving nucleophosmin (NPM) on 5q35 with a previously unidentified RTK on 2p23 (ALK), which occur in over 50% of anaplastic large cell lymphoma cases[1,6]. In contrast, about 5% of non-small cell lung cancer (NSCLC) cases generate recurrent ALK fusions with EML4, KIF5B, and HIP1[7,8,9].

Potential relevance: The first generation small molecule tyrosine kinase inhibitor (TKI), crizotinib[10], was FDA approved (2011) for the treatment of ALK positive advanced NSCLC. Kinase domain mutations including L1196M, G1269A, F1174L, G1202R, as well as other variants have been shown to confer acquired resistance to crizotinib in ALK positive NSCLC[11,12,13,14]. Other mechanisms of acquired resistance involve amplification of the ALK fusion gene and activation of alternate or bypass signaling pathways involving EGFR, KIT, MET, and IGF1R[15]. In order to overcome acquired resistance, second and third-generation ALK inhibitors including ceritinib[16] (2014),

## Biomarker Descriptions (continued)

alectinib[17] (2015), brigatinib[18] (2017), and lorlatinib[19] (2018) were developed and approved by the FDA. Two phase III trials evaluating crizotinib and alectinib as first line therapy in NSCLC, including patients with asymptomatic central nervous system (CNS) disease, were conducted and both studies showed consistent higher objective response rates (ORR) with alectinib relative to crizotinib[20,21]. For this reason, alectinib is a preferred first-line treatment of ALK positive NSCLC[22].

## Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

### EML4-ALK fusion

| Relevant Therapy | FDA | NCCN | EMA | ESMO | Clinical Trials* |
|---|---|---|---|---|---|
| ceritinib | ● | ◑ | ● | ● | ● (IV) |
| brigatinib | ● | ◑ | ● | ● | ✖ |
| crizotinib | ● | ◑ | ● | ● | ✖ |
| alectinib | ● | ● | ● | ● | ● (III) |
| lorlatinib | ● | ● | ● | ● | ✖ |
| next-generation ALK inhibitor | ✖ | ✖ | ✖ | ● | ✖ |
| brigatinib, alectinib | ✖ | ✖ | ✖ | ✖ | ● (III) |
| entrectinib | ✖ | ✖ | ✖ | ✖ | ● (II) |
| repotrectinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

## Relevant Therapy Details

### Current FDA Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

FDA information is current as of 2021-11-17. For the most up-to-date information, search www.fda.gov.

### EML4-ALK fusion

● **alectinib**

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-09-03 **Variant class:** ALK fusion or ALK overexpression

**Indications and usage:**
ALECENSA® is a kinase inhibitor indicated for the treatment of patients with anaplastic lymphoma kinase (ALK)-positive metastatic non-small cell lung cancer (NSCLC) as detected by an FDA-approved test.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/208434s012lbl.pdf

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.12(004).

## EML4-ALK fusion (continued)

### brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-09-24 **Variant class:** ALK fusion

**Indications and usage:**

ALUNBRIG® is a kinase inhibitor indicated for the treatment of adult patients with anaplastic lymphoma kinase (ALK)-positive metastatic non-small cell lung cancer (NSCLC) as detected by an FDA-approved test.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/208772s012lbl.pdf

### ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-10-07 **Variant class:** ALK fusion or ALK overexpression

**Indications and usage:**

ZYKADIA® is a kinase inhibitor indicated for the treatment of adults with metastatic non-small cell lung cancer (NSCLC) whose tumors are anaplastic lymphoma kinase (ALK)-positive as detected by an FDA-approved test.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/211225s004lbl.pdf

### crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-09-22 **Variant class:** ALK fusion

**Indications and usage:**

XALKORI® is a kinase inhibitor indicated for the treatment of

- patients with metastatic non-small cell lung cancer (NSCLC) whose tumors are anaplastic lymphoma kinase (ALK) or ROS1-positive as detected by an FDA-approved test.
- pediatric patients 1 year of age and older and young adults with relapsed or refractory, systemic anaplastic large cell lymphoma (ALCL) that is ALK-positive.
  - Limitations of Use: The safety and efficacy of XALKORI® have not been established in older adults with relapsed or refractory, systemic ALK-positive ALCL.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/202570s031lbl.pdf

### lorlatinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-03-03 **Variant class:** ALK fusion or ALK overexpression

**Indications and usage:**

LORBRENA® is a kinase inhibitor indicated for the treatment of adult patients with metastatic non-small cell lung cancer (NSCLC) whose tumors are anaplastic lymphoma kinase (ALK)-positive as detected by an FDA-approved test.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/210868s004lbl.pdf

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## Current NCCN Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

NCCN information is current as of 2021-11-01. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

## EML4-ALK fusion

### alectinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (First-line therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

### brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (First-line therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

### ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (First-line therapy); Other recommended intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

### crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (First-line therapy); Useful in certain circumstances

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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## EML4-ALK fusion (continued)

### lorlatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (First-line therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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### alectinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion or ALK overexpression

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Brain Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2021]

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### alectinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Metastatic, Advanced (Subsequent therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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### brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Brain Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2021]

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## EML4-ALK fusion (continued)

### brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Metastatic, Advanced (Subsequent therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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### ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Brain Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2021]

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### ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (Subsequent therapy)

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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### crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic (Subsequent therapy)

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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### lorlatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion or ALK overexpression

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Brain Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2021]

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## EML4-ALK fusion (continued)

### lorlatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Metastatic, Advanced (Subsequent therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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### crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**NCCN Recommendation category:** 2B

**Population segment (Line of therapy):**
- Brain Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2021]

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### brigatinib

**Cancer type:** Inflammatory Myofibroblastic Tumor **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- (Line of therapy not specified); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Soft Tissue Sarcoma [Version 2.2021]

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### ceritinib

**Cancer type:** Inflammatory Myofibroblastic Tumor **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- (Line of therapy not specified); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Soft Tissue Sarcoma [Version 2.2021]

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### crizotinib

**Cancer type:** Inflammatory Myofibroblastic Tumor **Variant class:** ALK fusion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- (Line of therapy not specified); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Soft Tissue Sarcoma [Version 2.2021]

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## Current EMA Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

EMA information is current as of 2021-11-17. For the most up-to-date information, search www.ema.europa.eu/ema.

## EML4-ALK fusion

### ● alectinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-10-11 **Variant class:** ALK fusion or ALK overexpression

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/alecensa-epar-product-information_en.pdf

### ● brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-10-15 **Variant class:** ALK fusion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/alunbrig-epar-product-information_en.pdf

### ● crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-07-21 **Variant class:** ALK fusion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/xalkori-epar-product-information_en.pdf

### ● ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-10-11 **Variant class:** ALK positive

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/zykadia-epar-product-information_en.pdf

### ● lorlatinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2021-11-05 **Variant class:** ALK positive

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/lorviqua-epar-product-information_en.pdf

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.12(004).

## Current ESMO Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

ESMO information is current as of 2021-11-01. For the most up-to-date information, search www.esmo.org.

## EML4-ALK fusion

### ● alectinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Stage IV (First-line therapy); ESMO-MCBS v1.1 score: 4
- Advanced, Progression (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 4

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### ● brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Stage IV (First-line therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### ● ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Advanced, Progression (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 4

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

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## EML4-ALK fusion (continued)

### crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Stage IV (First-line therapy); ESMO-MCBS v1.1 score: 4
- (Second-line therapy, Subsequent therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### next-generation ALK inhibitor

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Advanced (Second-line therapy, Subsequent therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / B

**Population segment (Line of therapy):**
- Stage IV (First-line therapy); ESMO-MCBS v1.1 score: 4

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### crizotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** I / B

**Population segment (Line of therapy):**
- Stage IV (First-line therapy); ESMO-MCBS v1.1 score: 4

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

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## EML4-ALK fusion (continued)

### next-generation ALK inhibitor

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** II / A

**Population segment (Line of therapy):**
- Advanced, Progression (Second-line therapy, Subsequent therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### alectinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** III / A

**Population segment (Line of therapy):**
- (First-line therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** III / A

**Population segment (Line of therapy):**
- Advanced (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 3

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### lorlatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** III / A

**Population segment (Line of therapy):**
- Stage IV; Advanced, Progression (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 3

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

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## EML4-ALK fusion (continued)

### brigatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** III / B

**Population segment (Line of therapy):**
- Advanced (First-line therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

### ceritinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**ESMO Level of Evidence/Grade of Recommendation:** IV / B

**Population segment (Line of therapy):**
- (First-line therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Metastatic Non-Small-Cell Lung Cancer [Online Guideline (15SEP2020 - https://www.esmo.org/guidelines/lung-and-chest-tumours/clinical-practice-living-guidelines-metastatic-non-small-cell-lung-cancer); Ann Oncol (2018) 29 (suppl 4): iv192–iv237.]

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## Clinical Trials in Taiwan region:

## Clinical Trials Summary

### EML4-ALK fusion

| NCT ID | Title | Phase |
|---|---|---|
| NCT03596866 | A Phase III Randomized Open-label Study of Brigatinib (Alunbrig) Versus Alectinib (Alecensa) in Advanced Anaplastic Lymphoma Kinase-Positive Non Small-Cell Lung Cancer Patients Who Have Progressed on Crizotinib (Xalkori) | III |
| NCT02584933 | An Open-label, Multi-center, Phase IV Roll-over Study in Patients With ALK Positive Malignancies Who Have Completed a Novartis-sponsored Ceritinib (LDK378) Study and Are Judged by the Investigator to Benefit From Continued Treatment With Ceritinib | IV |
| NCT03456076 | A Phase III, Open-Label, Randomized Study to Evaluate the Efficacy and Safety of Adjuvant Alectinib Versus Adjuvant Platinum-Based Chemotherapy in Patients With Completely Resected Stage IB (Tumors Equal to or Larger Than 4cm) to Stage IIIA Anaplastic Lymphoma Kinase Positive Non-Small Cell Lung Cancer | III |
| NCT02568267 | An Open-Label, Multicenter, Global Phase II Basket Study of Entrectinib for the Treatment of Patients With Locally Advanced or Metastatic Solid Tumors That Harbor NTRK1/2/3, ROS1, or ALK Gene Rearrangements. Studies of Tumor Alterations Responsive to Targeting Receptor Kinases (STARTRK-2) | II |
| NCT03093116 | A Phase I/II, Open-Label, Multi-Center, First-in-Human Study of the Safety, Tolerability, Pharmacokinetics, and Anti-Tumor Activity of TPX-0005 in Patients With Advanced Solid Tumors Harboring ALK, ROS1, or NTRK1-3 Rearrangements (TRIDENT-1) | I/II |
| NCT04094610 | A Phase I/II, Open-Label, Safety, Tolerability, Pharmacokinetics, and Anti-Tumor Activity Study of Repotrectinib in Pediatric and Young Adult Subjects With Advanced or Metastatic Malignancies Harboring ALK, ROS1, NTRK1-3 Alterations | I/II |

## Alerts Informed By Public Data Sources

### Current NCCN Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

NCCN information is current as of 2021-11-01. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

### EML4-ALK fusion

#### atezolizumab

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**Summary:**

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK fusions."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

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## EML4-ALK fusion (continued)

### nivolumab

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**Summary:**

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK fusions."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

### pembrolizumab

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** ALK fusion

**Summary:**

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK fusions."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 7.2021]

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.12(004).

## Signatures

**Testing Personnel:**

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